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Swiss Federal Institute of Technology Zurich

Oscillation-Free Discontinuous Galerkin Stabilization in MFEM

Master Thesis

Timo Zund

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Advisors: Dr. R. Käppeli

Seminar for Applied Mathematics,
Department of Mathematics, ETH Zürich

Abstract

This thesis studies maintainable oscillation-control mechanisms for discontinuous Galerkin discretizations of hyperbolic conservation laws in MFEM. It develops an adapted oscillation-free discontinuous Galerkin method with face-local characteristic speeds, restricts that method using the KXRCF troubled-cell detector, and implements the scale- and evolution-invariant 2024 OEDG method as a fixed comparator. Polynomial-shell damping is realized through finite-element projection operators, so the stored solution basis need not be modal, orthogonal, or Legendre-based. The implementation supports scalar equations and compressible Euler flow on distributed affine meshes and is accompanied by conservative admissibility treatment, an independent WENO5 reference solver, and deterministic serial and MPI tests.

The comparative numerical study is intentionally not finalized in this draft. Its proposed design separates smooth accuracy, discontinuity resolution, detector selectivity, invariance, robustness, computational cost, and parallel scaling. Final quantitative claims will be inserted only after the experiment matrix has been reviewed and an approved, fully traceable dataset has been generated. Reduced development calculations are retained solely to verify the reporting workflow and are not treated as thesis evidence.

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Chapter 1

Introduction

High-order discontinuous Galerkin (DG) methods combine element-local polynomial approximations with numerical fluxes at element interfaces. Their locality is attractive for parallel computing, but high-order polynomials can develop nonphysical oscillations near discontinuities. Oscillation-free DG methods address this by damping selected nonconstant modes while preserving cell averages [9, 8, 10].

The goal of this work is not only to reproduce a mathematical construction, but to integrate it into MFEM in a form that remains understandable, testable, and useful for several conservation laws. Four practical questions drive the implementation:

1. Which face propagation speed should scale the OFDG sensor for nonlinear systems such as Euler?
2. Can a troubled-cell indicator avoid unnecessary damping in smooth elements?
3. How should OFDG and the indicator exchange face data on a distributed mesh?
4. How does the adapted method compare with conventional RKDG and the scale-invariant 2024 OEDG method under reproducible accuracy, robustness, and cost measurements?

1.1 Related work and research gap

The RKDG framework combines element-local DG spatial discretizations with strong-stability-preserving explicit Runge–Kutta methods [2, 4]. Near discontinuities, nonlinear limiters commonly use a troubled-cell indicator to avoid modifying smooth regions. KXRCE uses DG interface superconvergence for this purpose [6]; systematic comparisons and later alternatives

show that detector selectivity remains sensitive to normalization, polynomial degree, and the chosen indicator variable [11, 3, 14, 15].

OFDG instead adds projection-based damping whose strength is inferred from interelement jumps [9, 8]. The 2024 OEDG method recasts that damping as exact modal attenuation and introduces scale- and evolution-invariant coefficients [10]. The unresolved practical question addressed here is how these ingredients behave when implemented in a general finite element library, combined with selective KXRCF activation, and extended across distributed meshes. The combined implementation is not one of the published methods verbatim; its deviations and empirical validation must therefore be stated explicitly.

1.2 Contributions

The first contribution is a publicly accessible, reusable MFEM realization of adapted OFDG and a fixed 2024 OEDG comparator. Its polynomial-shell damping is assembled through L^2 projection matrices in the finite-element basis supplied by MFEM. The represented solution therefore does not need to use modal, orthogonal, or Legendre coefficients. The same construction can be transferred to another polynomial DG framework using its standard basis evaluation, first-derivative, quadrature, transformation, and degree-of-freedom operations. Higher derivatives required by the sensors are generated internally by repeated application of first-derivative coefficient operators.

The second contribution is the coupling of adapted OFDG to a multidimensional, multicomponent KXRCF activation mask. An inactive element is copied unchanged by the damping step, recovering the unfiltered DG update there and avoiding unnecessary smooth-region attenuation. Whether this selectivity improves global error or end-to-end runtime is a question for the final measurements, not an assumption of the method.

The software contribution further includes a shared face-physics interface, Euler normal spectral radii $|v \cdot n| + c$, cached element and face operators, MPI shared-face processing, conservative Euler positivity limiting, and reusable MFEM example drivers. Tests cover frozen numerical fingerprints, conservation, vector orderings, multiple polynomial bases, affine triangle and quadrilateral geometries, analytical wave speeds, scale invariance, admissibility, and a discontinuity located exactly on a two-rank partition boundary. A manifest-driven pipeline stores raw outputs and profiles, computes derived metrics, and generates the report figures and tables without manual transcription.

1.3 Organization

Chapter 2 defines RKDG, adapted OFDG, KXRCE, and the 2024 OEDG comparator, followed by the common numerical controls, software realization, parallelization, and experimental methodology. Chapter 3 presents numerical observations, Chapter 4 interprets their scientific meaning and limitations, and Chapter 5 summarizes the answers and future work. In the present draft, numerical observations are explicitly preliminary because the final experiment matrix has not yet been approved.

Chapter 2

Methods

This chapter defines the numerical methods before discussing their software realization or empirical behavior. The presentation begins with the common RKDG baseline, then introduces adapted OFDG, KXRCE, and the independent 2024 OEDG comparator in that order. Positivity treatment and reference solutions are common experimental controls rather than parts of any one stabilization method.

2.1 Governing equations and conventional RKDG

Consider a system of hyperbolic conservation laws

$$\partial_t \mathbf{u} + \nabla \cdot \mathbf{f}(\mathbf{u}) = 0 \quad \text{in } \Omega \times (0, T], \quad \mathbf{u}(\mathbf{x}, 0) = \mathbf{u}_0(\mathbf{x}). \quad (2.1)$$

Scalar advection and Burgers' equation use one component. For compressible Euler in d dimensions,

$$\mathbf{u} = (\rho, \rho \mathbf{v}, E)^\top, \quad p = (\gamma - 1) \left(E - \frac{1}{2} \rho |\mathbf{v}|^2 \right), \quad (2.2)$$

and the flux in direction j is

$$\mathbf{f}_j(\mathbf{u}) = (\rho v_j, \rho v_j v + p e_j, (E + p) v_j)^\top. \quad (2.3)$$

Let V_h^k be the discontinuous polynomial space on $\mathcal{T}_h$. The conventional semidiscrete DG method seeks $\mathbf{u}_h(t) \in [V_h^k]^m$ such that, on every element K ,

$$\int_K \partial_t \mathbf{u}_h \cdot \mathbf{w}_h \, dx - \int_K \mathbf{f}(\mathbf{u}_h) : \nabla \mathbf{w}_h \, dx + \int_{\partial K} \widehat{\mathbf{f}}(\mathbf{u}_h^-, \mathbf{u}_h^+, \mathbf{n}_K) \cdot \mathbf{w}_h^- \, ds = 0 \quad (2.4)$$

for all $\mathbf{w}_h \in [V_h^k]^m$. The numerical experiments use the same Rusanov-type interface flux for all compared methods. Explicit Runge–Kutta integration produces the RKDG baseline; OFDG and OEDG alter only the post-stage or post-step state, not Eq. (2.4). This separation follows the standard RKDG framework for nonlinear hyperbolic systems [2, 4].

2.2 Adapted Oscillation-Free Discontinuous Galerkin Stabilization

This chapter describes the oscillation-free discontinuous Galerkin (OFDG) stabilization used in this work. The presentation deliberately separates the mathematical operator from its realization in MFEM. The construction combines the projection-based damping mechanism introduced for scalar conservation laws in [9] with the scale normalization, wave-speed scaling, and exact damping evolution developed in [10]. The extension of the original OFDG method to hyperbolic systems is described in [8]. The precise multidimensional sensor implemented here differs from the sensors in those papers; the differences are stated explicitly in Section 2.2.2.

2.2.1 Mathematical formulation

Purpose of the stabilization

Consider a hyperbolic conservation law, or more specifically the transport problem

$$\partial_t u + \nabla \cdot f(u) = 0, \quad \text{or} \quad \partial_t u + \mathbf{a} \cdot \nabla u = 0. \quad (2.5)$$

Here $u = u(\mathbf{x}, t)$ is the transported scalar quantity, $\mathbf{x} \in \Omega$ is position, t is time, $f(u)$ is its physical flux, and $\mathbf{a}$ is a prescribed transport velocity. The first equation is the conservative form; the second is the special case of linear advection. A system of conservation laws replaces u by a vector of components. The OFDG construction below is applied to each component of that vector.

A discontinuous Galerkin method approximates the solution by a separate polynomial on every mesh element. The polynomials on two neighboring elements are not required to agree at their common face. This is useful because it makes the method local and permits discontinuous physical solutions, but a high-order polynomial can develop nonphysical over- and undershoots close to a shock or another unresolved feature.

OFDG adds damping only to the nonconstant polynomial content of the numerical solution. The amount of damping is determined automatically from jumps between neighboring element polynomials. A smooth, well-resolved solution has small jumps and therefore receives little damping. A discontinuity produces large jumps and therefore stronger damping. The construction has three ingredients:

1. local L^2 projections separate low- and high-order polynomial content;
2. jumps of the solution and its physical derivatives measure a lack of smoothness across element faces;

3. nonnegative damping rates combine the jump measure, element size, solution scale, and normal transport speed.

The following subsections define each ingredient before combining them into the stabilization operator.

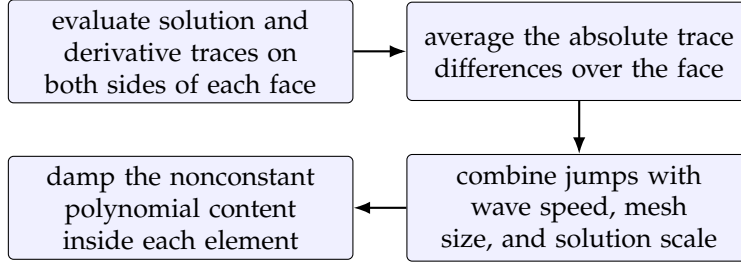


Figure 2.1: Conceptual OFDG pipeline. Information is gathered at element faces, converted into element-local damping rates, and finally applied through local polynomial projections.

Discrete setting

Let $\Omega \subset \mathbb{R}^d$ be partitioned into a mesh $\mathcal{T}_h$, and let $\mathcal{F}_h^i$ denote the set of interior faces. On each element $K \in \mathcal{T}_h$, let $V^k(K)$ be the polynomial space of degree $k \geq 1$ used by the DG discretization. The discontinuous finite element space is

$$V_h^k = \left\{ v \in L^2(\Omega) : v|_K \in V^k(K) \text{ for every } K \in \mathcal{T}_h \right\}. \quad (2.6)$$

The symbol d is the spatial dimension, K denotes one mesh element, and the subscript h indicates that the space is tied to a mesh with a characteristic element size h . An interior face is a full side shared by two elements; an outer boundary side is not included in $\mathcal{F}_h^i$. The notation $v|_K$ means the restriction of v to K . Because that restriction is chosen independently on every element, a function in V_h^k may have two different values at an interior face.

For a system with m components, the numerical solution belongs to the vector-valued space

$$\mathbf{u}_h = (u_{h,1}, \dots, u_{h,m})^\top \in [V_h^k]^m. \quad (2.7)$$

The present implementation treats these components independently. It does not perform a characteristic decomposition.

On an element K , choose a basis $\{\varphi_i\}_{i=1}^{N_k}$ of $V^k(K)$. A local DG polynomial is represented by

$$u_h|_K(\mathbf{x}) = \sum_{i=1}^{N_k} u_{K,i} \varphi_i(\mathbf{x}). \quad (2.8)$$

The numbers $u_{K,i}$ are the local degrees of freedom. Depending on the basis, they may be nodal values or general expansion coefficients. No assumption is made here that $\{\varphi_i\}$ is modal, orthogonal, or formed from Legendre polynomials. The word *modal* used later describes the invariant polynomial-degree subspaces selected by L^2 projections; it does not prescribe the coefficient basis in which the DG framework stores u_h . Both the theory and its matrix realization below are independent of that storage basis.

For $0 \leq r \leq k$, denote by $\Pi_K^r : L^2(K) \rightarrow V^r(K)$ the element-local L^2 projection,

$$(\Pi_K^r w - w, v)_K = 0 \quad \text{for every } v \in V^r(K), \quad (2.9)$$

Here $(p, q)_K = \int_K pq \, dx$ is the usual elementwise L^2 inner product. Equation (2.9) says that the projection error is orthogonal to every polynomial of degree at most r . Equivalently, among all functions in $V^r(K)$, $\Pi_K^r w$ is the one closest to w in the root-mean-square sense over K .

We adopt the convention $\Pi_K^{-1} = \Pi_K^0$. The high-pass operators used by the stabilization are

$$\mathcal{R}_K^{(l)} = I - \Pi_K^{l-1}, \quad 0 \leq l \leq k. \quad (2.10)$$

In particular, $\mathcal{R}_K^{(0)} = \mathcal{R}_K^{(1)} = I - \Pi_K^0$. Every $\mathcal{R}_K^{(l)} w$ has zero cell average.

The projection $\Pi_K^r w$ is the best approximation to w in $V^r(K)$ in the $L^2(K)$ norm. It retains polynomial content up to degree r and removes the higher-order content. Consequently, $\mathcal{R}_K^{(l)} w = w - \Pi_K^{l-1} w$ contains precisely the part that the l th damping contribution is allowed to suppress. The convention $\Pi_K^{-1} = \Pi_K^0$ ensures that no damping contribution changes the cell average, which is the constant mode carrying the locally conserved quantity.

Face jumps of physical derivatives

For the multi-index $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}_0^d$, define

$$|\alpha| = \sum_{j=1}^d \alpha_j, \quad \partial^\alpha = \partial_{x_1}^{\alpha_1} \cdots \partial_{x_d}^{\alpha_d}. \quad (2.11)$$

The entries of α count how often differentiation is taken in each coordinate direction, while $|\alpha|$ is the total derivative order. In two dimensions, for example, $\alpha = (1, 0)$ represents ∂_x , $\alpha = (0, 1)$ represents ∂_y , and $\alpha = (1, 1)$ represents ∂_{xy} .

Let $F = \partial K^- \cap \partial K^+$ be an interior face. The sign convention for the jump is immaterial below because only its absolute value occurs; we set

$$[[w]]_F = w^-|_F - w^+|_F. \quad (2.12)$$

The superscripts $-$ and $+$ are labels for the two neighboring elements, not signs. Thus $w^-|_F$ means: approach F from inside K^- and evaluate the polynomial stored on K^- . The trace $w^+|_F$ is obtained independently from K^+ . Even though both traces refer to the same physical points on F , they need not be equal in a DG method.

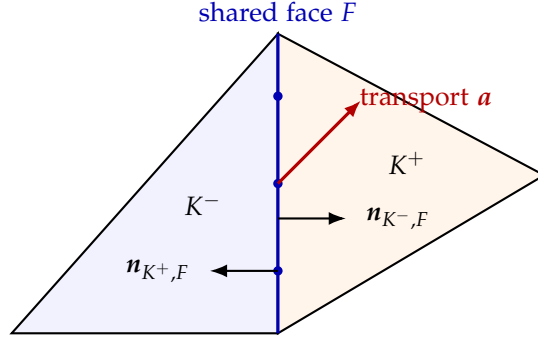


Figure 2.2: Two triangular DG elements sharing the interior face F . The dots indicate representative face quadrature points. At each dot the two element polynomials supply separate traces. Their difference forms the jump. The outward normals point in opposite directions, but the absolute normal wave speed is the same for a single-valued velocity.

For component c and total derivative order l , the implemented face sensor is the sum of normalized mean absolute jumps

$$J_{F,c}^{(l)} = \sum_{|\alpha|=l} \frac{1}{|F|} \int_F |\llbracket \partial^\alpha u_{h,c} \rrbracket_F| \, ds. \quad (2.13)$$

Every symbol in this expression has a direct interpretation:

- F is the shared face shown in Fig. 2.2, and $|F|$ is its length in two dimensions or area in three dimensions;
- $c \in \{1, \dots, m\}$ selects one physical solution component;
- $l \in \{0, \dots, k\}$ is the total derivative order;
- the sum includes every coordinate derivative ∂^α whose total order is l ;
- $\llbracket \partial^\alpha u_{h,c} \rrbracket_F$ is the difference between the two independently evaluated derivative traces; and
- the integral averages the magnitude of that difference over the entire face after division by $|F|$.

Thus $J_{F,c}^{(0)}$ measures the jump of the DG solution itself, while the higher values of l measure all physical partial derivatives of total order l . For example, in two dimensions the $l = 1$ contribution contains the jumps of $\partial_x u_h$ and $\partial_y u_h$, while the $l = 2$ contribution contains those of $\partial_{xx} u_h$, $\partial_{xy} u_h$,

and $\partial_{yy}u_h$. The absolute value is applied pointwise before integration, so positive and negative jumps cannot cancel. The sum over all multi-indices ensures that the sensor does not select only one coordinate direction.

Dividing by $|F|$ turns the face integral into a face average. Hence two faces with the same typical jump magnitude produce comparable values even if their physical measures differ. In one spatial dimension a face is a point and (2.13) reduces to the ordinary absolute jump. An earlier version accumulated squared jumps and subsequently took a square root. The current definition uses the intended absolute jump directly; it is both simpler and cheaper to evaluate.

As a simple scalar example, suppose the traces are constant on F with $u_h^-|_F = 2$ and $u_h^+|_F = 1.25$. Then $J_F^{(0)} = |2 - 1.25| = 0.75$, independently of the length of F . If the difference varies along the face, the quadrature described later approximates its mean absolute value instead. This distinction matters: averaging the signed difference could hide a nonsmooth face when positive and negative differences cancel, whereas averaging its absolute value cannot.

Let $\mathbf{a}(\mathbf{x})$ be the prescribed transport velocity and $\mathbf{n}_{K,F}$ the outward unit normal of K on F . The normal wave speed used on the K side of F is

$$\beta_{K,F} = \max_{\mathbf{x} \in F} |\mathbf{a}(\mathbf{x}) \cdot \mathbf{n}_{K,F}|. \quad (2.14)$$

The dot product extracts the part of the velocity crossing the face. Tangential transport has zero normal component and should not determine the exchange between the two neighboring elements. The absolute value makes the rate independent of which side supplies the orientation of the normal. The maximum chooses a safe representative speed when the velocity varies along the face. In the code this maximum is approximated by evaluating the velocity at the face quadrature points. When no velocity is supplied, the implementation uses $\beta_{K,F} = 1$.

Global scale and element damping rates

Following the scale-invariant idea of [10], each component is normalized by one global amplitude. Define the componentwise extrema

$$u_{\min,c} = \min_{K \in \mathcal{T}_h} \min_{\mathbf{x} \in K} u_{h,c}(\mathbf{x}), \quad u_{\max,c} = \max_{K \in \mathcal{T}_h} \max_{\mathbf{x} \in K} u_{h,c}(\mathbf{x}). \quad (2.15)$$

The global mean and amplitude are

$$\bar{u}_{h,c} = \frac{1}{|\Omega|} \int_{\Omega} u_{h,c} \, d\mathbf{x}, \quad (2.16)$$

$$A_c = \max \{ \bar{u}_{h,c} - u_{\min,c}, u_{\max,c} - \bar{u}_{h,c} \} = \|u_{h,c} - \bar{u}_{h,c}\|_{L^\infty(\Omega)}. \quad (2.17)$$

2.2. Adapted Oscillation-Free Discontinuous Galerkin Stabilization

If $A_c = 0$, all damping rates for component c are defined to be zero. For $A_c > 0$, introduce

$$C_{k,l} = \frac{2l+1}{2(2k-1)l!}, \quad 0 \leq l \leq k. \quad (2.18)$$

The damping rate accumulated for element K , component c , and derivative order l is

$$\delta_{K,c}^{(l)}(\mathbf{u}_h) = C_{k,l} \frac{h_K^{l-1}}{A_c} \sum_{\substack{F \subset \partial K \\ F \in \mathcal{F}_h^i}} \beta_{K,F} J_{F,c}^{(l)}. \quad (2.19)$$

For convenient reference, the complete definition is collected below as one system:

$$\begin{aligned} \beta_{K,F} &= \max_{\mathbf{x} \in F} |\mathbf{a}(\mathbf{x}) \cdot \mathbf{n}_{K,F}|, \\ \delta_{K,c}^{(l)}(\mathbf{u}_h) &= \frac{2l+1}{2(2k-1)} \frac{h_K^{l-1}}{l!} \sum_{\substack{F \subset \partial K \\ F \in \mathcal{F}_h^i}} \beta_{K,F} \frac{\sum_{|\alpha|=l} \frac{1}{|F|} \int_F |[\![\partial^\alpha u_{h,c}]\!]_F| \, ds}{\|u_{h,c} - \bar{u}_{h,c}\|_{L^\infty(\Omega)}}. \end{aligned} \quad (2.20)$$

The symbols can be read as follows:

- K is the element that receives damping, and ∂K is its boundary;
- the face sum runs only over interior faces of K , because only those faces possess two DG traces;
- k is the polynomial degree of the DG space and $l \in \{0, \dots, k\}$ is the derivative order whose jump is being measured;
- c selects one solution component;
- h_K is a characteristic length of element K ; and
- $\mathbf{a} \cdot \mathbf{n}_{K,F}$ is the component of the transport velocity that crosses the face.

The factor h_K^{l-1} consists of the h_K^l derivative scaling of the sensor and the additional h_K^{-1} rate in the damping equation. Consequently, $\delta_{K,c}^{(l)}$ has the dimensions of an inverse time. Equation (2.19) is the exact coefficient computed by the current code; in the implementation it is stored under the name `sigma_elem`.

The terms in (2.19) have distinct roles. The face sum collects information from every interior neighbor of K . The jump $J_{F,c}^{(l)}$ has the physical units of the solution divided by length to the power l . Multiplication by h_K^{l-1} and by the speed $\beta_{K,F}$ converts it to the units of the solution divided by time. Division by A_c leaves an inverse-time damping rate. A larger jump, a faster

wave, or a smaller element therefore increases the rate at which unresolved polynomial content is removed.

For a concrete scalar example, take degree $k = 2$, derivative order $l = 0$, one contributing face, $h_K = 0.1$, $A = 1$, normal speed $\beta = 2$, and mean absolute solution jump $J = 0.3$. Since $C_{2,0} = 1/[2(2 \cdot 2 - 1)] = 1/6$, the resulting contribution is

$$\delta_K^{(0)} = \frac{1}{6} \frac{0.1^{-1}}{1} (2)(0.3) = 1. \quad (2.21)$$

If time is measured in seconds, this is a damping rate of one per second. A second interior face would contribute another nonnegative term to the same rate.

The normalization makes the rates invariant under a rescaling and shift of a component. More precisely, away from the zero-amplitude threshold, replacing $u_{h,c}$ by $\lambda u_{h,c} + b$, with $\lambda \neq 0$, leaves every $\delta_{K,c}^{(l)}$ unchanged. Multiplying the transport velocity by a positive factor multiplies the rates by the same factor, so the product of a damping time and a damping rate is unchanged under the corresponding rescaling of physical time.

Projection-based damping operator

The local damping operator is

$$\mathcal{D}_{K,c}(\mathbf{u}_h)u_{h,c} = \sum_{l=0}^k \delta_{K,c}^{(l)}(\mathbf{u}_h) \mathcal{R}_K^{(l)} u_{h,c}. \quad (2.22)$$

Substituting the definition of the high-pass operator gives the same formula in a form that exposes exactly what is removed from the element polynomial:

$$\mathcal{D}_{K,c}(\mathbf{u}_h)u_{h,c} = \sum_{l=0}^k \delta_{K,c}^{(l)}(\mathbf{u}_h) \left(u_{h,c}|_K - \Pi_K^{l-1}(u_{h,c}|_K) \right), \quad \Pi_K^{-1} := \Pi_K^0. \quad (2.23)$$

For each l , the projection reconstructs the part of the element polynomial of degree at most $l - 1$. Subtracting it leaves only higher-degree content. That remainder is multiplied by the nonnegative rate associated with derivative jumps of order l . Finally, all orders are added. The operation is entirely local once the face-based rates have been computed: no value from a neighboring element is needed while applying (2.23).

When OFDG is used as a semidiscrete stabilization, the ordinary DG operator is augmented by

$$(\partial_t u_{h,c}, v_h)_K = \mathcal{A}_{K,c}(\mathbf{u}_h; v_h) - (\mathcal{D}_{K,c}(\mathbf{u}_h)u_{h,c}, v_h)_{K'}, \quad (2.24)$$

where $\mathcal{A}_{K,c}$ denotes the unstabilized DG spatial discretization. The function v_h is an arbitrary DG test function, and $(p, q)_K = \int_K pq \, dx$. The minus sign

in front of the OFDG term is essential: because the operator is nonnegative, subtracting it removes energy from nonconstant modes instead of amplifying them. The implementation returns the coefficient vector of $\mathcal{D}_{K,c}u_{h,c}$; the caller subtracts this vector from the DG time derivative.

To see which polynomial content is affected, consider one term in (2.22). The $l = 0$ and $l = 1$ terms both act on $u_{h,c} - \Pi_K^0 u_{h,c}$, namely everything except the cell average. For $l \geq 2$, the l th term acts only on content not representable by polynomials of degree $l - 1$. Thus a derivative jump of order l is connected to damping of polynomial modes of degree at least l . The total operator is nonlinear because the rates are recomputed from the current solution, although it is a linear combination of projections once those rates have been frozen.

For example, when $k = 3$, the action of the four rates is:

rate	remainder being multiplied	polynomial content affected
$\delta_{K,c}^{(0)}$	$u_{h,c} - \Pi_K^0 u_{h,c}$	degrees 1, 2, and 3
$\delta_{K,c}^{(1)}$	$u_{h,c} - \Pi_K^0 u_{h,c}$	degrees 1, 2, and 3
$\delta_{K,c}^{(2)}$	$u_{h,c} - \Pi_K^1 u_{h,c}$	degrees 2 and 3
$\delta_{K,c}^{(3)}$	$u_{h,c} - \Pi_K^2 u_{h,c}$	degree 3 only

The constant, degree-zero part never appears in a remainder and is therefore never damped. This is the algebraic reason that the element average is preserved.

The damping term is locally conservative because

$$\int_K \mathcal{R}_K^{(l)} w \, dx = \int_K \left(w - \Pi_K^{l-1} w \right) dx = 0 \quad \text{for all } l. \quad (2.25)$$

For $l = 0$, the already stated convention $\Pi_K^{-1} = \Pi_K^0$ is used. The integral vanishes because an L^2 projection onto a space containing constants preserves the element integral of the projected function.

It is also dissipative. Orthogonality of the L^2 projection and nonnegativity of the rates give

$$(\mathcal{D}_{K,c} u_{h,c}, u_{h,c})_K = \sum_{l=0}^k \delta_{K,c}^{(l)} \left\| u_{h,c} - \Pi_K^{l-1} u_{h,c} \right\|_{L^2(K)}^2 \geq 0. \quad (2.26)$$

These two identities are direct consequences of the implemented projection operator and do not depend on the particular face sensor.

Exact decay by polynomial-degree shells

The same damping operator can be applied in a separate pseudo-time step. Freeze the rates at an input state $\mathbf{u}_h$ and solve, independently on each element and for each component,

$$\frac{\partial w_{K,c}}{\partial s} + \sum_{l=0}^k \delta_{K,c}^{(l)}(\mathbf{u}_h) \mathcal{R}_K^{(l)} w_{K,c} = 0, \quad w_{K,c}(0) = u_{h,c}|_K. \quad (2.27)$$

Define the mutually orthogonal polynomial shells

$$\mathcal{Q}_K^{(0)} = \Pi_K^0, \quad \mathcal{Q}_K^{(j)} = \Pi_K^j - \Pi_K^{j-1}, \quad 1 \leq j \leq k, \quad (2.28)$$

and their cumulative damping rates

$$\Lambda_{K,c}^{(j)} = \sum_{l=0}^j \delta_{K,c}^{(l)}, \quad 1 \leq j \leq k. \quad (2.29)$$

Since a mode in the j th shell is contained in $\mathcal{R}_K^{(l)}$ precisely when $l \leq j$, the exact solution at pseudo-time τ is

$$w_{K,c}(\tau) = \Pi_K^0 u_{h,c} + \sum_{j=1}^k \exp\left(-\tau \Lambda_{K,c}^{(j)}\right) \mathcal{Q}_K^{(j)} u_{h,c}. \quad (2.30)$$

Eliminating the abbreviations $\Lambda_{K,c}^{(j)}$ and $\mathcal{Q}_K^{(j)}$ gives the directly usable form

$$w_{K,c}(\tau) = \Pi_K^0 u_{h,c} + \sum_{j=1}^k \exp\left(-\tau \sum_{l=0}^j \delta_{K,c}^{(l)}(\mathbf{u}_h)\right) \left(\Pi_K^j - \Pi_K^{j-1}\right) u_{h,c}. \quad (2.31)$$

Here j labels a polynomial shell, $\tau \geq 0$ is the duration of the damping step, and the inner sum is the total rate acting on that shell. The exponential factor lies between zero and one: it equals one if the accumulated rate is zero and approaches zero when the rate or damping time becomes large. The first term $\Pi_K^0 u_{h,c}$ is the cell-average mode and has no exponential factor, so it is copied unchanged.

Thus the cell average is unchanged, whereas increasingly high polynomial shells are multiplied by nonnegative decay factors. This is the modal-filter interpretation emphasized in [10]. It is a statement about the orthogonal decomposition of the polynomial space, not about the basis used to store the coefficients: every projector and shell in Eq. (2.30) is applied directly in the host basis. The exact map avoids a time-step restriction caused solely by stiffness of the frozen damping equation.

The coefficient-space stabilization and the exact decay are two representations of the same operator. Indeed,

$$\sum_{l=0}^k \delta_{K,c}^{(l)} \mathcal{R}_K^{(l)} = \sum_{j=1}^k \Lambda_{K,c}^{(j)} \mathcal{Q}_K^{(j)}. \quad (2.32)$$

The first form is used by `ComputeStabilization`; the exponential of the second form is used by `CompDecay`.

2.2.2 Implementation decisions and details

Scope and geometric assumptions

The implementation is a standalone MFEM header. It assumes that the finite element space is nonempty and discontinuous, that $k \geq 1$, and that all elements have the same geometry, order, dimension, number of local degrees of freedom, and component layout. For a multicomponent field, the local degrees of freedom of each component must form one contiguous block. All element maps are assumed to be affine, and the mesh must remain unchanged during the lifetime of the OFDG object. These conditions are preconditions rather than properties checked at run time.

Different elements need not have the same Jacobian. They may have different sizes and may be translated, rotated, or affinely skewed. General bilinear quadrilaterals and curved elements are outside the present scope because their inverse Jacobians vary within an element.

More explicitly, MFEM describes a physical element by a map from a standard reference element. The affine assumption means that this map has the form

$$\mathbf{x} = T_K(\boldsymbol{\xi}) = \mathbf{b}_K + J_K \boldsymbol{\xi}, \quad (2.33)$$

where $\boldsymbol{\xi}$ is a reference coordinate, $\mathbf{b}_K$ is a translation, and the matrix J_K is constant within K . The matrix may differ from one element to another. Its scaling, rotation, and shear describe the size, orientation, and skew of the physical element. What the implementation requires is therefore a constant Jacobian on each individual element, not one globally identical Jacobian for the whole mesh. The characteristic length h_K in the damping rate is obtained from MFEM's element-size routine for that element.

Projection matrices

Let $\{\varphi_i\}_{i=1}^{N_k}$ be the local degree- k basis and $\{\psi_a^{(r)}\}_{a=1}^{N_r}$ the degree- r basis. The code assembles

$$(M_k)_{ij} = (\varphi_i, \varphi_j)_K, \quad (M_r)_{ab} = (\psi_a^{(r)}, \psi_b^{(r)})_K, \quad (2.34)$$

$$(B_r)_{ai} = (\psi_a^{(r)}, \varphi_i)_K. \quad (2.35)$$

The matrix representation of $I - \Pi_k^r$ in the degree- k coefficient basis is

$$Q_r = M_k - B_r^\top M_r^{-1} B_r, \quad S_r = M_k^{-1} Q_r = I - M_k^{-1} B_r^\top M_r^{-1} B_r. \quad (2.36)$$

Thus applying S_r to a degree- k coefficient vector directly returns the coefficients of the remainder $w - \Pi_k^r w$. The matrices M_k and M_r are the degree- k and degree- r mass matrices, respectively; B_r couples the two bases; and I is the $N_k \times N_k$ identity matrix. The code stores S_r for $0 \leq r \leq k$.

Crucially, this construction never converts the solution to Legendre or other orthogonal modal coefficients. It represents the same projection operator in whatever supported scalar polynomial DG basis the host framework supplies. Consequently, a nodal basis, a modal basis, or another polynomial coefficient basis produces the same function-space operation up to quadrature and roundoff. This basis-agnostic realization is what makes the implementation portable: a second framework need not add a special OFDG solution basis, provided it can form the corresponding polynomial-space projections.

On an affine element, all three mass and coupling matrices contain the same constant physical Jacobian factor, which cancels in S_r . Therefore the projection matrices are assembled on the first element and reused on every element even when the element Jacobians differ.

Coefficient-to-coefficient derivatives

The repeated derivatives required by (2.13) are represented as polynomial coefficients rather than only as values at quadrature points. At the finite element interpolation points $\{\xi_q\}_{q=1}^{N_k}$, define

$$V_{qi} = \varphi_i(\xi_q), \quad G_{qi}^{(r)} = \frac{\partial \varphi_i}{\partial \xi_r}(\xi_q). \quad (2.37)$$

If $\mathbf{u}$ is a coefficient vector, $G^{(r)} \mathbf{u}$ contains reference-derivative values at the interpolation points. The coefficient vector of that derivative is therefore

$$\mathbf{u}_{\xi_r} = D_{\xi_r} \mathbf{u}, \quad D_{\xi_r} = V^{-1} G^{(r)}. \quad (2.38)$$

This Vandermonde construction is kept explicit in the code. For the affine map $\mathbf{x} = T_K(\xi)$, the physical derivative matrices are

$$D_{x_j}^K = \sum_{r=1}^d (J_K^{-1})_{rj} D_{\xi_r}. \quad (2.39)$$

The inverse Jacobian J_K^{-1} is read once at the reference-element centre and one set of matrices $\{D_{x_j}^K\}_{j=1}^d$ is stored for every element.

Only first reference derivatives of the supplied finite-element basis are requested from MFEM. Repeated application of the resulting coefficient operators generates every higher derivative needed by OFDG. In framework terms, the construction requires scalar polynomial DG basis values and first derivatives, quadrature, nested lower-order polynomial spaces or equivalent projection operators, element and face transformations, and local degree-of-freedom access. It therefore avoids a modal-basis requirement, but it does not claim support for arbitrary nonpolynomial, $H(\text{div})$, $H(\text{curl})$, or mixed finite-element spaces.

All multi-indices with total degree at most k are organized in a directed acyclic graph. Each nonzero multi-index selects one parent of degree one less and one physical derivative direction. Walking this graph evaluates every required derivative once. The number of stored derivative states per element and component is

$$\sum_{l=0}^k \binom{l+d-1}{d-1} = \binom{k+d}{d}. \quad (2.40)$$

Face quadrature and cached evaluation

MFEM remains the sole source of mesh connectivity. The setup loops over MFEM face numbers and obtains the two adjacent elements from MFEM's face-transformation data. Boundary faces are skipped. For each interior face, a quadrature rule of order $2k+1$ is selected for the MFEM face geometry. The following data are cached directly at the MFEM face index:

$$(E_F^-)_{qi} = \varphi_i(\xi_{F,q}^-), \quad (E_F^+)_{qi} = \varphi_i(\xi_{F,q}^+), \quad \hat{\omega}_q = \frac{\omega_q}{\sum_p \omega_p}. \quad (2.41)$$

For affine faces the physical face Jacobian is constant and cancels between the integral and $|F|$ in (2.13); hence the normalized reference weights $\hat{\omega}_q$ are sufficient. All derivative traces and all components on one side of a face are evaluated together in one dense matrix multiplication. There is no face-pattern matching or deduplication.

If $\mathbf{u}_\alpha^-$ and $\mathbf{u}_\alpha^+$ are the coefficient vectors of the same physical derivative on the two neighboring elements, the actual quadrature calculation is

$$J_{F,c}^{(l)} \approx \sum_{|\alpha|=l} \sum_{q=1}^{N_{q,F}} \hat{\omega}_q \left| (E_F^- \mathbf{u}_\alpha^-)_q - (E_F^+ \mathbf{u}_\alpha^+)_q \right|. \quad (2.42)$$

The implementation therefore performs one subtraction and one absolute value for each derivative, component, and quadrature point. No squared-jump accumulator or final square root is required.

Runtime sequence and storage policy

For every input state, the implementation performs the following operations:

1. evaluate the global mean and scale (2.17) at volume quadrature points;
2. rebuild and store all derivative coefficient states on every element;
3. loop over MFEM interior faces, evaluate the normal wave speeds and the mean absolute jump measures (2.13), and accumulate the rates (2.19) on both neighboring elements;
4. apply either the coefficient-space damping operator (2.22) or the exact decay map (2.30).

The volume mean is integrated with a rule of order $2k + 1$. The extrema used for A_c are the minimum and maximum over the same quadrature values. In floating point arithmetic, the implementation sets the rates of component c to zero when $A_c \leq 10^{-14}$.

There is no bounded or dynamic derivative cache. Every element owns permanent storage for its derivative coefficients, and all of it is rebuilt once for the current state. Likewise, each interior face owns its two evaluation matrices and normalized quadrature weights. Ignoring small container overheads, the principal storage costs are

$$\mathcal{O}(|\mathcal{T}_h| dN_k^2) \quad \text{for physical derivative operators,} \quad (2.43)$$

$$\mathcal{O}\left(|\mathcal{T}_h| mN_k \binom{k+d}{d}\right) \quad \text{for derivative coefficient states,} \quad (2.44)$$

$$\mathcal{O}\left(\sum_{F \in \mathcal{F}_h^i} N_{q,F} (2N_k + 1)\right) \quad \text{for face evaluation data.} \quad (2.45)$$

This policy uses more predictable memory than an evicting cache and removes cache lookups and replacement decisions from the face loop.

An optional active-element mask can restrict stabilization or decay to selected elements. A face is omitted only when neither neighbor is active, so an active element still receives contributions from every adjacent interior face. Runtime scratch arrays and optional internal timers are owned by the OFDG object. Because that scratch state is mutable and reused, concurrent calls on the same object are not supported.

Relation to the published formulations

The original multidimensional OFDG sensors in [9, 8] measure derivative jumps at element vertices. The system formulation in [8] additionally trans-

forms jumps to characteristic variables and takes a maximum over characteristic components. The current implementation instead uses the face-averaged absolute-jump measure (2.13) and constructs a separate rate for each physical component. Its global amplitude normalization, normal wave-speed factor, and exact decay map are motivated by the later OEDG formulation [10].

Consequently, the implementation should be regarded as the specific hybrid OFDG and OEDG damping variant defined by Eq. (2.13), Eq. (2.19), and Eq. (2.22), not as a literal reproduction of one published algorithm. Conservation and dissipation follow directly from Eq. (2.25) and Eq. (2.26); convergence results proved for the published vertex-based or characteristic-variable sensors do not, by themselves, constitute a proof for this modified sensor.

The KXRCF troubled-cell indicator supplies the optional active-element set used by this mask interface. Its definition, normalization, and system coupling are given in Section 2.3.

2.3 KXRCF Troubled-Cell Indication

The KXRCF detector was introduced by Krivodonova, Xin, Remacle, Chevaugnon, and Flaherty for discontinuous Galerkin approximations of hyperbolic conservation laws [6]. In contrast to a limiter or to OFDG, a troubled-cell indicator does not modify the numerical solution. It supplies a binary decision about where a separate stabilization mechanism should act. This distinction is important in the present method: KXRCF selects elements and the adapted OFDG operator defined in Section 2.2 damps their nonconstant polynomial content.

2.3.1 Mathematical motivation

Interface superconvergence and jump detection

Let $K \in \mathcal{T}_h$ be an element with outward unit normal $\mathbf{n}_K$. The DG approximation has independent traces on the two sides of an interior face. If N denotes the neighboring element, define the trace jump viewed from K by

$$[u_h]_{K,N} = u_h|_K - u_h|_N. \quad (2.46)$$

For a smooth exact solution, the two DG traces approximate the same physical value and their jump decreases rapidly under mesh refinement. The original KXRCF construction exploits the particularly accurate DG trace on an outflow boundary: at an inflow face of K , the neighboring trace is an outflow trace of the upstream element. A jump that is too large relative to the expected smooth-mesh behavior therefore signals that at least one adjacent

polynomial does not represent a smooth solution. This superconvergence-based motivation also explains the connection between KXRCF and the superconvergence measurements used later in the numerical study [6, 11].

Let $\boldsymbol{v}$ denote the transport velocity used solely for the inflow/outflow decision. The inflow part of the element boundary is

$$\partial K^- = \{\boldsymbol{x} \in \partial K : \boldsymbol{v}(\boldsymbol{x}) \cdot \boldsymbol{n}_K(\boldsymbol{x}) < 0\}. \quad (2.47)$$

Faces for which the normal velocity vanishes make no contribution. This directional definition differs from sensors that inspect all faces symmetrically: KXRCF deliberately measures information entering K .

Normalized indicator

Let k be the polynomial degree, h_K an element radius, S_K a scale of the solution on K , and $|\partial K^-|$ the measure of the inflow boundary. For a scalar DG solution, the KXRCF indicator used in this work is

$$I_K = \frac{\left| \int_{\partial K^-} [u_h]_{K,N} \, ds \right|}{h_K^{(k+1)/2} |\partial K^-| S_K}. \quad (2.48)$$

The element is marked troubled when

$$I_K > \tau, \quad \tau = 1 \text{ by default}. \quad (2.49)$$

The face measure turns the numerator into an averaged signed jump, S_K removes the local solution magnitude, and the factor $h_K^{(k+1)/2}$ separates the expected smooth and discontinuous regimes. If the face-averaged jump is $\mathcal{O}(h_K^{k+1})$ in a smooth region and $S_K = \mathcal{O}(1)$, then

$$I_K = \mathcal{O}\left(h_K^{(k+1)/2}\right) \longrightarrow 0. \quad (2.50)$$

If an unresolved discontinuity produces an $\mathcal{O}(1)$ jump, the same normalization instead gives

$$I_K = \mathcal{O}\left(h_K^{-(k+1)/2}\right), \quad (2.51)$$

which grows under refinement. These estimates explain the mesh-independent threshold concept; they are an asymptotic motivation, not a new proof for the modified multicomponent detector implemented here. Comparative studies show that detector behavior can still depend on polynomial degree, the chosen indicator variable, and the threshold [11, 3, 14, 15].

One-dimensional specialization

In one dimension, ∂K^- is one endpoint whenever the transport velocity has a definite sign. For $K = (x_{j-1/2}, x_{j+1/2})$, positive velocity, and $h_K = \Delta x_j/2$, Eq. (2.48) becomes

$$I_j = \frac{|u_h(x_{j-1/2}^+) - u_h(x_{j-1/2}^-)|}{(\Delta x_j/2)^{(k+1)/2} |\bar{u}_j|}. \quad (2.52)$$

The code uses the absolute cell average for the one-dimensional scale, matching the common presentation of the detector. In two and three dimensions it uses the maximum absolute value at the element quadrature points. The different normalizations follow the original multidimensional use of a local maximum norm rather than treating point boundaries as positive-measure faces.

Signed integration and cancellation

The absolute value in Eq. (2.48) is applied after the signed jump has been integrated over all inflow faces. Thus

$$\left| \int_{\partial K^-} [u_h]_{K,N} \, ds \right| \neq \int_{\partial K^-} |[u_h]_{K,N}| \, ds \quad \text{in general.} \quad (2.53)$$

The first expression is the KXRCF construction and permits cancellation along or between inflow faces. The second is an L^1 jump sensor of the type used by the 2024 OEDG comparator. Keeping these definitions distinct prevents the implementation from silently turning KXRCF into a different detector.

2.3.2 Hyperbolic systems and project-specific choices

For component c of a conservative state, the implementation forms

$$I_{K,c} = \frac{\left| \int_{\partial K^-} (u_{K,c} - u_{N,c}) \, ds \right|}{h_K^{(k+1)/2} |\partial K^-| S_{K,c}}, \quad I_K = \max_c I_{K,c}. \quad (2.54)$$

One active set is then shared by the complete conservative state,

$$\mathcal{A} = \{K \in \mathcal{T}_h : I_K > \tau\}. \quad (2.55)$$

This pooling prevents density, momentum, and energy from being filtered on different cells. It is a project-specific extension: published Euler uses of KXRCF commonly monitor selected variables such as density, entropy, or total energy rather than the maximum over every conservative component [6, 3].

For scalar advection, the prescribed velocity determines inflow. Burgers' equation uses the secant velocity

$$v_n = \frac{1}{2}(u_1 + u_2) \mathbf{n}, \quad (2.56)$$

and Euler uses the arithmetic mean of the two fluid velocities,

$$v_n = \frac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2) \cdot \mathbf{n}. \quad (2.57)$$

The latter is an advective direction, not the Euler spectral radius. Subsonic Euler interfaces contain characteristic fields traveling in both directions, so no single scalar velocity represents every characteristic family. OFDG and OEDG therefore use the full one-sided radii $|\mathbf{v}_i \cdot \mathbf{n}| + c_i$ for damping strength, whereas Eq. (2.57) is used only to decide which element receives the KXRCF inflow contribution. A characteristic-variable KXRCF detector remains outside the present scope.

2.3.3 Coupling to adapted OFDG

After a completed Runge–Kutta step, KXRCF computes the active set Eq. (2.55). A face is evaluated whenever at least one adjacent element is active, ensuring that every active element receives all of its face contributions. The exact OFDG decay is then applied only to active elements. Because the OFDG decay leaves the constant polynomial mode unchanged, this gating does not alter cell-average conservation. KXRCF therefore changes where damping is applied, but neither its modal form nor its conservation argument.

2.3.4 Implementation decisions

Geometry, quadrature, and normalization

In one dimension h_K is half the physical element length. In higher dimensions the implementation maps the reference-element center and vertices and takes the largest center-to-vertex distance. This is an exact circumradius for the supported symmetric cells and a documented generalized radius for other affine MFEM element families. Volume quadrature supplies $S_{K,c}$; face quadrature integrates the signed trace jump and the physical inflow measure.

A relative scale floor prevents division by a vanishing component scale. It is defined from the global maximum of the component scales and is used only when the local scale is smaller. A globally constant zero component has zero jump and can never mark an element. Physical boundary faces are omitted because no neighboring DG trace exists; boundary conditions remain the responsibility of the hyperbolic DG operator.

Distributed ownership and caches

Element evaluation matrices, quadrature weights, radii, local-face trace matrices, and normals are immutable after construction. During an indicator application, ordinary interior faces update both adjacent elements. An MPI shared face updates only the locally owned element after MFEM exchanges the neighbor coefficients; the neighboring rank performs the complementary update. All ranks participate in the component-scale reductions before any root-only diagnostics are printed.

The first implementation recomputed radii, shapes, normals, and quadrature weights on every call. On the fixed benchmark with 9,216 quadrilateral elements, degree three, four components, and 100 indicator calls, this required 2.8784 seconds. The cached version requires 1.9436 seconds, a 32.5% reduction. Face integration remains the dominant phase at 1.5875 seconds, followed by element scales at 0.3372 seconds and normalization at 0.0141 seconds. Further compression was not adopted because it would obscure the geometry and ownership logic for a comparatively small additional benefit.

2.3.5 Limitations

The implementation assumes affine, full-dimensional MFEM elements and a fixed mesh during the lifetime of the indicator. Its radius generalization, component pooling, scale floor, boundary omission, and Euler inflow velocity are engineering choices rather than claims from the original KXRCF analysis. Moreover, selecting fewer cells does not itself guarantee that the resulting OFDG–KXRCF method is nonoscillatory or positivity preserving. These questions must be evaluated empirically and are therefore represented explicitly in the proposed experiment matrix.

2.4 The 2024 OEDG Comparator

The numerical comparison requires an implementation of the method published by Peng, Sun, and Wu in 2024 [10] which is independent of the project-specific decision to gate OFDG with KXRCF. This chapter states the implemented choices explicitly. In particular, the 2024 method is not a “vertex-only” method: its multidimensional sensor is an interface integral of absolute jumps. Vertex evaluation belongs to the earlier multidimensional OFDG construction [9, 8].

2.4.1 Scale-invariant interface sensor

Let K be an element, $e \subset \partial K$ a face, k the DG degree, and $m \in \{0, \dots, k\}$. For component i , define the global amplitude

$$A_i = \left\| u_h^{(i)} - \text{avg}_\Omega(u_h^{(i)}) \right\|_{L^\infty(\Omega)}. \quad (2.58)$$

The maximum is evaluated at the same volume quadrature nodes used by the filter. When A_i is below a relative machine-scale tolerance, all associated sensors are zero. Otherwise the face contribution corresponding to equation (3.10) of [10] is

$$\sigma_{e,K}^{m,(i)} = \frac{(2m+1)h_{e,K}^m}{2(2k-1)m! A_i} \sum_{|\alpha|=m} \frac{1}{|e|} \int_e \left| \llbracket \partial^\alpha u_h^{(i)} \rrbracket \right| dS, \quad (2.59)$$

where $h_{e,K} = \sup_{x \in K} \text{dist}(x, e)$. The implementation computes this face height geometrically and caches it; on a simplex it is the altitude associated with e , not a generic element diameter. In 2D the face average uses the endpoint trapezoidal rule employed in the paper. Point faces in 1D require no quadrature choice. The present 3D path retains a proper high-order face rule because the paper specifies the trapezoidal treatment only for 2D computations.

For a system, the paper avoids characteristic decomposition and pools physical components,

$$\sigma_{e,K}^m = \max_i \sigma_{e,K}^{m,(i)}. \quad (2.60)$$

The damping rate combines all adjacent faces,

$$\delta_K^m = \sum_{e \subset \partial K} \frac{\beta_e}{h_{e,K}} \sigma_{e,K}^m, \quad (2.61)$$

where β_e is the local normal spectral radius. Thus advection uses $|\mathbf{a} \cdot \mathbf{n}|$, Burgers uses the corresponding normal flux derivative, and Euler uses $|\mathbf{v} \cdot \mathbf{n}| + c$. This is the material correction to the historical unit-speed assumption.

Multiplying a full conservative state by a positive constant multiplies both the jump numerator and A_i by that constant, leaving Eq. (2.59) unchanged. Adding a spatially constant state also leaves derivative jumps and deviations from the global mean unchanged. These are the scale- and evolution-invariance properties tested directly by the unit suite and by the scaled Lax experiment.

2.4.2 Exact attenuation without a prescribed coefficient basis

Let P^m denote the element-local L^2 projection into degree at most m , with $P^{-1} = P^0$. During the OE pseudo-time problem, the coefficients Eq. (2.61)

are frozen at the candidate RK stage. The published derivation may be written using coefficients of a local orthogonal basis, with scaled Legendre polynomials as one convenient one-dimensional choice. Such a basis is not mathematically required. In basis-independent operator form, the exact attenuation is

$$\mathcal{F}_{\tau,K}u_h = P^0u_h + \sum_{j=1}^k \exp\left(-\tau \sum_{m=0}^j \delta_K^m\right) (P^j - P^{j-1})u_h, \quad (2.62)$$

so the constant subspace is unchanged and every degree- j polynomial shell is scaled by its corresponding exponential. The implementation assembles the projectors in MFEM's supplied coefficient basis and applies Eq. (2.62) without ever forming Legendre coefficients. No pseudo-time discretization error is introduced, and every element mean is preserved to roundoff.

The public OEDG2024 type fixes the face quadrature, face-height, component-pooling, normalization, wave-speed, and exact-decay choices. It reuses the already tested projection, derivative, immutable face cache, and MPI halo machinery of the adapted OFDG kernel. This sharing removes duplicate geometry code without allowing the comparator to inherit KXRCF or an experiment-only option accidentally. It also gives OFDG and OEDG the same basis-agnostic projection realization, preventing a change of coefficient basis from becoming an uncontrolled difference in the numerical comparison.

2.4.3 Runge–Kutta cadence and comparison contract

The published OEDG algorithm applies the OE map after every Runge–Kutta stage. Consequently, `-method oedg -cadence auto` selects the small stage driver. For the project OFDG method, `auto` uses an MFEM solver and one filter application after a complete step. The explicit stage and step choices are retained for cadence ablations. The headline comparison therefore consists of:

1. conventional RKDG;
2. adapted OFDG, applied post-step only in KXRCF cells;
3. 2024 OEDG, applied after every RK stage.

Adapted OFDG without KXRCF is reported as an ablation. The thesis uses these names only for these fixed configurations.

2.5 Common Numerical Controls

2.5.1 Euler admissibility

The common positivity limiter preserves each conservative cell average and scales all local degrees of freedom toward that mean. Density is constrained first. Pressure is then constrained by bisection along the same line segment, using the concavity of the admissible Euler set. The evaluation set contains volume quadrature nodes and every local or shared face point used by the DG and filter operators. This construction follows the mean-preserving scaling principle used by positivity-preserving high-order DG schemes [17].

If a mean is itself nonphysical, no mean-preserving correction exists. The candidate stage is rejected and the step is retried with half the time step up to the documented retry limit. Assisted comparisons apply the same limiter to RKDG, adapted OFDG, and OEDG. Unassisted experiments record failure time and minimum density and pressure separately; they are not silently discarded.

2.5.2 Independent one-dimensional reference

An independent finite-volume program supplies discontinuous one-dimensional reference solutions. It uses component-wise fifth-order WENO reconstruction, local Lax–Friedrichs flux splitting, and SSPRK3 [5, 4]. It shares no DG projection, indicator, or filter code. Validation uses a smooth periodic Euler wave and the exact Sod Riemann solution [13]. Before a final reference is accepted, two successive reference grids must differ by substantially less than the error of every compared DG solution.

2.6 Implementation and Parallelization

2.6.1 Software structure

The implementation separates physics from geometry. A *face physics* object receives two conservative traces and a unit normal, and returns the two nonnegative propagation radii required by OFDG together with the signed transport velocity required by KXRCE. Policies are provided for unit-speed compatibility, prescribed advection, Burgers flow, and compressible Euler. This keeps the OFDG and KXRCE loops independent of equation-specific indexing.

The implementation also separates a function-space operation from its coefficient representation. Polynomial-shell projections and first-derivative maps are assembled once in the scalar DG basis supplied by MFEM. Run-time filtering then consists of dense operations on the existing local coeffi-

cients; neither the examples nor another host framework must store their solution in a Legendre or otherwise modal basis.

The example programs reuse MFEM's `AdvectionFlux`, `BurgersFlux`, `EulerFlux`, `RusanovFlux`, `HyperbolicFormIntegrator`, and explicit ODE solvers. Adapted OFDG is applied once after a complete MFEM step. Earlier cadence experiments found no accuracy improvement from applying it at all four RK4 stages, so the custom stage-wise RK4 remains a regression experiment rather than the production OFDG path. A separate small experiment driver implements SSPRK3 and RK4 callbacks for paper-faithful stage-wise OEDG and for Euler positivity checks. Rejected callbacks leave both the accepted state and physical time unchanged.

The command-line method and cadence interface is common to all drivers. The auto cadence resolves to post-step MFEM integration for DG and OFDG, and to stage-wise integration for OEDG. Compatibility flags map onto the same enumeration; they do not create a second implementation path.

All maintained example drivers provide the same optional live-visualization interface. Passing `-vis` streams the current parallel grid function to GLVis, while `-vs` selects the update interval. A shared wrapper owns the socket protocol and requires a successful connection on every MPI rank; otherwise all ranks continue with visualization disabled. Visualization is off by default and is not part of measured study runs.

2.6.2 Distributed meshes

MFEM distinguishes faces whose two elements are owned by one rank from shared faces whose second element is a halo neighbor. Each stabilization call follows the same sequence:

1. compute local element data and reduce global component scales;
2. exchange face-neighbor DG coefficients;
3. process ordinary interior faces and update both local neighbors;
4. process shared faces and update only the locally owned element;
5. apply the element-local OFDG decay map.

The neighboring rank independently performs the complementary update. KXRFCF uses the same ownership rule. A two-rank test partitions a one-dimensional mesh so that its only discontinuity lies on the processor boundary; it checks that KXRFCF marks the inflow cell, OFDG and OEDG observe the derivative jump, and every element mean remains unchanged. Collective diagnostic reductions are called by all ranks before root-only printing. This ordering is protected by the same integration test.

2.6.3 Verification strategy

The test suite was established before changing the face physics. It includes the following layers:

- exact frozen OFDG stabilization and decay vectors for a discontinuous one-dimensional state;
- conservation, inactive-mask behavior, uniform-polynomial invariance, and affine triangle and quadrilateral geometries;
- KXRCE zero, constant, one- and two-dimensional jump fingerprints, scale invariance, component pooling, and MFEM vector-order invariance;
- analytical Burgers and Euler normal speeds and rejection of negative density or pressure;
- OEDG constants, means, modal attenuation, scale/evolution invariance, and one- and two-dimensional fingerprints;
- positivity invariance, mean preservation, positive evaluation states, nonphysical means, and MPI consistency;
- the MPI partition-boundary regression, stage rejection, RK cadence, reference-solver, rate-calculation, and reduced end-to-end study tests.

Debug tests use address and undefined-behavior sanitizers. Release tests are compiled separately so that stale debug objects cannot be mistaken for an optimized build.

2.7 Experimental Methodology

The numerical study is currently in its design phase. The final manifest is therefore marked *draft* and cannot be executed by the study runner. A dry run expands every proposed row, checks its metadata, and estimates its cost from development measurements without launching a numerical solver. Final measurements will begin only after the matrix has been reviewed and its status is changed explicitly to *approved*.

2.7.1 Comparison contract

The headline methods are conventional RKDG, adapted OFDG restricted by KXRCE, and the 2024 OEDG comparator. All-cell adapted OFDG is a targeted ablation. For a given row, the methods share the governing problem, mesh, polynomial degree, numerical flux, CFL rule, final time, and, where required, the common positivity treatment. Comparisons across degrees are reported both at equal mesh and at approximately equal actual global degrees of freedom; these answer different questions and are not combined.

Smooth problems record L^1 , L^2 , and L^∞ errors. For uniform refinement from h to $h/2$, the observed rate is

$$r = \frac{\log(E_h/E_{h/2})}{\log 2}. \quad (2.63)$$

The 1D advection superconvergence experiment additionally records the cell-average and downwind-point errors defined in Section 3.1. Discontinuous cases record independent reference error, total variation, extrema, overshoot, undershoot, conservation drift, troubled- and limited-cell activity, rejected steps, and failure time.

Timing is separate from accuracy. Proposed timing rows specify one unrecorded warm-up and five repetitions, to be summarized by the median and interquartile range. Fixed-work strong scaling uses one, two, and four MPI ranks. A pilot will select a workload whose one-rank time is long enough to dominate startup cost. No timing value from the reduced CI checkpoint is interpreted as a performance result.

2.7.2 Proposed experiment families

The approval matrix includes smooth P^1 – P^3 sequences for one- and two-dimensional advection, pre-shock Burgers flow, the one- and two-dimensional Euler waves, the transported vortex, and a triangular-mesh sequence. The discontinuous group contains post-shock Burgers flow, scaled Lax data, Woodward–Colella blast waves [16], the Shu–Osher shock–entropy interaction [12], and a two-dimensional Euler Riemann problem [7].

Targeted ablations vary the KXRCF threshold, compare all-cell and gated OFDG, compare physical and unit wave-speed scaling, and compare stage-wise with post-step OFDG cadence. Robustness is measured first without assistance and then with the common positivity limiter. The exact proposed meshes, final times, metrics, references, estimated costs, and intended report artifacts are listed in Section A.1; they remain proposals rather than completed experiments.

2.7.3 Reproducibility and provenance

Each eventual row will record its full command, actual DOFs, time-integration parameters, MPI ranks, source revision, MFEM revision, compiler, host, and UTC timestamp. Raw logs and rank-local profiles are retained. Derived CSV, tables, and figures are generated in a separate deterministic analysis step. The software implementation uses MFEM for meshes, finite element spaces, parallel ownership, and operator assembly [1].

Chapter 3

Results

Preliminary development results—not for final comparison.
The values in this chapter come from the reduced workflow used to test data collection and report generation. Meshes, final times, repetitions, and the experiment matrix have not yet been approved for the thesis-final study. The figures and tables demonstrate presentation style only and must not be cited as final numerical evidence.

3.1 Convergence-table convention

For the periodic one-dimensional advection problem, define

$$e_1 = \|u - u_h\|_{L^1(\Omega)}, \quad (3.1)$$

$$e_2 = \|u - u_h\|_{L^2(\Omega)}, \quad (3.2)$$

$$e_3 = \|u - u_h\|_{L^\infty(\Omega)}, \quad (3.3)$$

$$e_4 = \left(h \sum_j \left| \frac{1}{h} \int_{I_j} (u - u_h) dx \right|^2 \right)^{1/2}, \quad (3.4)$$

$$e_5 = \left(h \sum_j \left| u(x_{j+1/2}) - u_h(x_{j+1/2}^-) \right|^2 \right)^{1/2}. \quad (3.5)$$

The first three errors measure the full polynomial solution. The latter two measure cell-average and downwind-point superconvergence after Gauss–Radau initialization. Each method receives a separate table with identical degree and mesh rows. A missing first rate is written as “–” rather than assigned a numerical value.

3. RESULTS

Table 3.1: Preliminary development run: errors and convergence rates for RKDG on smooth one-dimensional advection.

k	N_x	e_1	rate	e_2	rate	e_3	rate	e_4	rate	e_5	rate
1	20	7.04e-03	–	8.26e-03	–	1.58e-02	–	1.35e-03	–	1.78e-03	–
	40	1.77e-03	1.99	2.11e-03	1.97	4.47e-03	1.82	1.81e-04	2.90	2.36e-04	2.92
	80	4.36e-04	2.02	5.30e-04	1.99	1.17e-03	1.94	2.34e-05	2.96	3.01e-05	2.97
2	20	3.28e-04	–	4.26e-04	–	9.33e-04	–	7.19e-06	–	1.07e-05	–
	40	4.00e-05	3.04	5.35e-05	2.99	1.18e-04	2.98	2.35e-07	4.94	3.50e-07	4.93
	80	4.96e-06	3.01	6.69e-06	3.00	1.48e-05	3.00	7.46e-09	4.98	1.10e-08	4.99

Table 3.2: Preliminary development run: errors and convergence rates for adapted OFDG–KXRCF on smooth one-dimensional advection.

k	N_x	e_1	rate	e_2	rate	e_3	rate	e_4	rate	e_5	rate
1	20	1.37e-02	–	1.76e-02	–	3.18e-02	–	1.28e-02	–	1.90e-02	–
	40	2.62e-03	2.39	3.24e-03	2.44	7.71e-03	2.04	2.25e-03	2.50	2.71e-03	2.81
	80	5.27e-04	2.31	6.55e-04	2.30	2.18e-03	1.82	3.64e-04	2.63	4.12e-04	2.72
2	20	6.90e-04	–	9.73e-04	–	2.18e-03	–	8.00e-04	–	9.00e-04	–
	40	4.47e-05	3.95	5.82e-05	4.06	1.57e-04	3.80	2.13e-05	5.23	1.63e-05	5.79
	80	4.99e-06	3.16	6.69e-06	3.12	1.48e-05	3.41	2.06e-07	6.69	3.70e-07	5.46

Table 3.3: Preliminary development run: errors and convergence rates for 2024 OEDG on smooth one-dimensional advection.

k	N_x	e_1	rate	e_2	rate	e_3	rate	e_4	rate	e_5	rate
1	20	1.47e-02	–	1.80e-02	–	3.15e-02	–	1.44e-02	–	1.76e-02	–
	40	2.89e-03	2.34	3.52e-03	2.36	6.54e-03	2.27	2.66e-03	2.44	3.02e-03	2.55
	80	5.84e-04	2.31	6.74e-04	2.39	1.15e-03	2.50	4.03e-04	2.72	4.39e-04	2.78
2	20	1.05e-03	–	1.25e-03	–	2.85e-03	–	1.01e-03	–	1.30e-03	–
	40	6.68e-05	3.97	8.47e-05	3.88	2.01e-04	3.83	5.37e-05	4.23	6.45e-05	4.33
	80	6.12e-06	3.45	7.85e-06	3.43	1.83e-05	3.46	2.84e-06	4.24	3.37e-06	4.26

3.2 Smooth accuracy presentation

Convergence curves are separated by polynomial degree so that an axis contains only the three headline methods and one reference slope. Complete values remain in the corresponding paper-style tables.

3.3 Discontinuity and invariance presentation

One-dimensional discontinuity figures show at most the three headline methods and the independent reference. A matched inset is retained only when it reveals a steep-gradient difference that is not visible in the full-domain view. Scale-invariance figures use one panel per method and

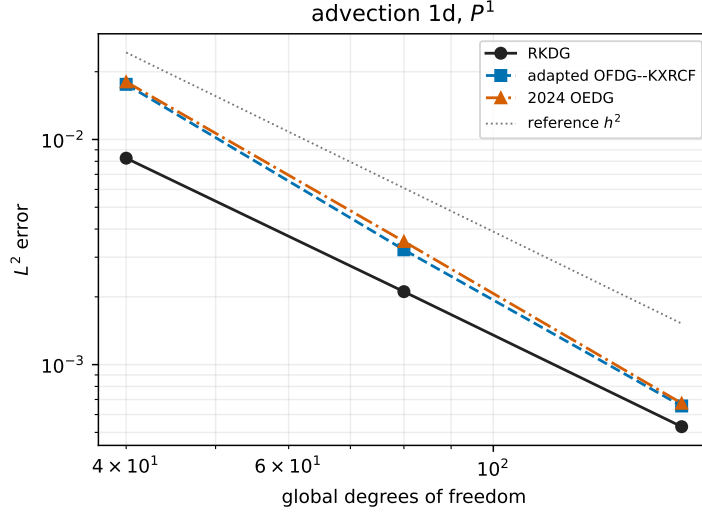


Figure 3.1: Preliminary development run for periodic 1D advection with P^1 elements. The plot compares L^2 error against actual global degrees of freedom for the three headline methods; the dotted line is the expected second-order slope. These values are not part of the final study.

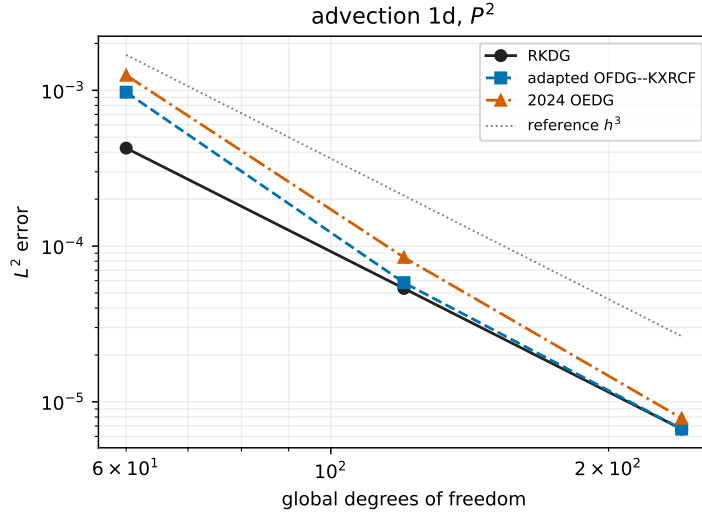


Figure 3.2: Preliminary development run for periodic 1D advection with P^2 elements. The plot compares L^2 error against actual global degrees of freedom for the three headline methods; the dotted line is the expected third-order slope. These values are not part of the final study.

3. RESULTS

compare normalized solutions at fixed numerical settings.

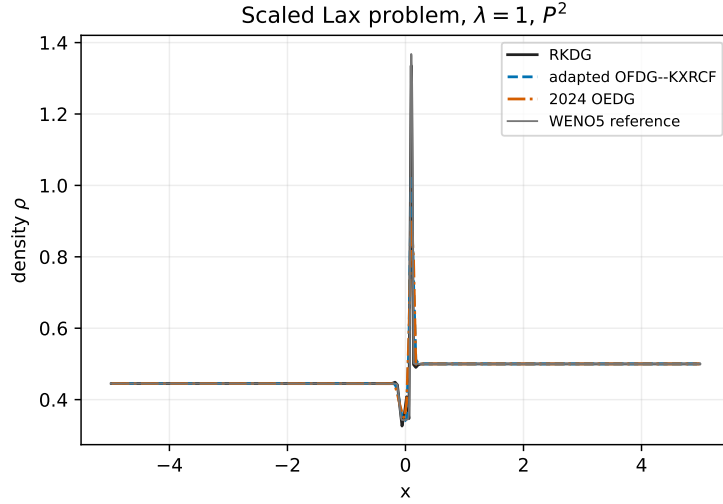


Figure 3.3: Preliminary development profile for scaled Lax data with P^2 elements at the reduced final time. Density is compared with the independent WENO5 reference. These values are not part of the final study.

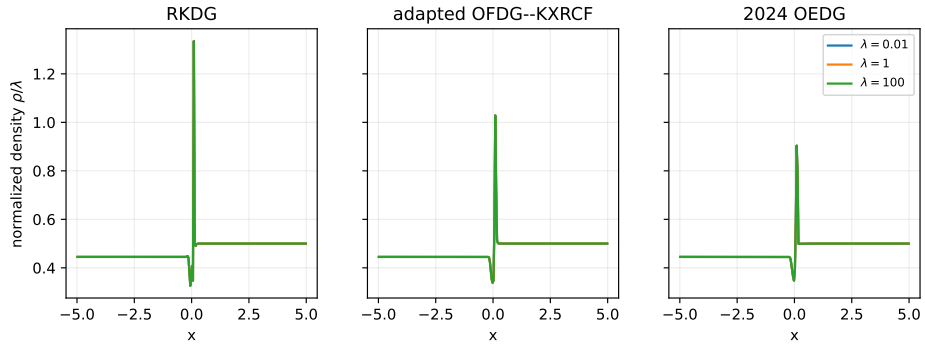


Figure 3.4: Preliminary normalized-density presentation for the three proposed Lax scales with P^2 elements. Each panel contains one method, so coincident curves directly test scale invariance without mixing methods. These values are not part of the final study.

3.4 Two-dimensional presentation

Contour comparisons and line cuts are deliberately separated. The contour figure uses identical limits and levels for all methods; the line-cut figure then has sufficient width for direct quantitative comparison.

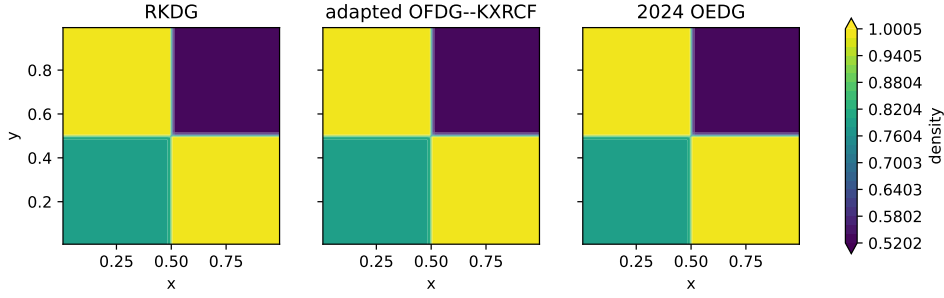


Figure 3.5: Preliminary development density contours for the reduced two-rank 2D Riemann problem with P^1 elements. Identical contour levels are used for all headline methods. These values are not part of the final study.

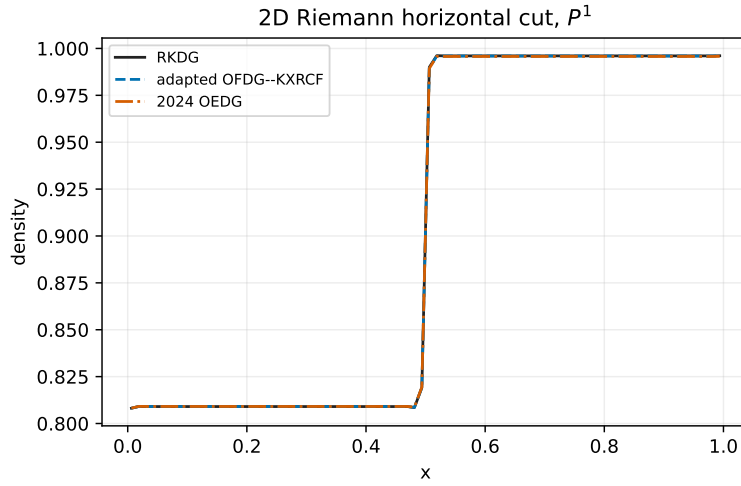


Figure 3.6: Horizontal density cuts corresponding to Fig. 3.5. The sampling convention is identical for all methods. These values are not part of the final study.

3.5 Detector, robustness, and cost presentation

The final KXRCF result will show indicator values, the threshold, and active cells along the same spatial coordinate as a representative shock profile. The reduced checkpoint did not record spatial indicator values, so no surrogate activation chart is shown here. Positivity activity and unassisted failure times will be tabulated rather than compressed into a many-category bar chart. Runtime plots will be problem- and degree-specific and will use repeated timing rows only; the single-run development timings are deliberately omitted.

Chapter 4

Discussion

This chapter is intentionally limited to the interpretation framework. The final discussion will be written only after the approved experiment matrix has been executed. It will answer the research questions in the order posed in Chapter 2: the effect of physical wave-speed scaling, the selectivity and cost of KXRCF gating, distributed-memory behavior, and the comparative accuracy and robustness of RKDG, adapted OFDG–KXRCF, and OEDG.

The discussion will distinguish observations from established theory. In particular, conservation follows from preservation of cell averages, whereas convergence, nonoscillatory behavior, and robustness of the modified sensor must be supported by the final measurements. It will also address temporal error, reference-grid independence, positivity-limiter intervention, detector threshold sensitivity, implementation-specific bases and fluxes, and the limited rank count of the local scaling study. Basis independence will be distinguished from geometric generality: the projection construction does not prescribe a coefficient basis, whereas the current verified implementation still assumes fixed, affine, uniform-order scalar polynomial DG elements.

Chapter 5

Conclusion

The software and verification framework are complete, but the comparative experiment design is still under review. Consequently, this draft makes no final ranking of RKDG, adapted OFDG–KXRCF, and OEDG. The final conclusion will state only findings supported by the approved dataset and will identify remaining theoretical and computational work, including characteristic KXRCF variants, curved meshes, larger-rank scaling, and paper-scale two-dimensional robustness cases.

Appendix A

Reproducibility and Study Design

From the repository root, the draft workflow is

```
make test
make test-study
make study-quick
make figures-draft
make report-draft
```

The first command builds sanitizer-enabled serial tests and runs the two-rank partition test. The second validates analysis code and the independent WENO solver. The reduced study exists to exercise the complete data path; its outputs are not final numerical evidence.

The proposed full design is inspected without running solvers using

```
make study-plan.
```

This produces an expanded CSV and an estimated runtime summary. Because the full manifest is marked draft, `make study-full` exits before building or launching a solver. Execution becomes possible only after the matrix has been reviewed and its manifest status is changed to approved.

Raw logs, rank-local sampled profiles, and the long-form CSV are written below `measurements/study/<profile>`. Draft report assets are mirrored under `report/generated/draft`; approved final assets use the separate `report/generated/final` directory. Each numerical row records its exact command, source revision, compiler, MFEM revision, host, and UTC timestamp. Numerical failures remain rows with their exit code, diagnostic, and detected failure time.

Live GLVis output can be enabled for the advection, Burgers, and Euler drivers with `-vis`; `-vs` selects the timestep interval between updates. Euler can additionally write rank-local MFEM solution files with `-save`. These

paths are intended for interactive verification only; every thesis plot is generated headlessly from recorded CSV or profile data.

A.1 Proposed experiment matrix

The machine-readable manifest is the source of truth. The following compact view records why each family is proposed and where it is intended to appear. Every row remains unapproved in this draft.

Family	Scientific question	Intended artifact	Status
1D advection	Optimal P^1 – P^3 convergence and Gauss–Radau superconvergence	Separate method tables and degree-specific curves	proposed
Smooth Burgers	Accuracy for a nonlinear scalar flux before shock formation	Three method tables	proposed
Smooth Euler wave	Accuracy for a hyperbolic system and density-error behavior	Three method tables	proposed
2D advection	Dimensional extension on Cartesian meshes	Degree-specific curves and tables	proposed
Triangular advection	Sensitivity to a non-tensor affine element family	Supplementary convergence tables	proposed
Euler density wave	Two-dimensional system convergence	Density tables	proposed
Transported vortex	Smooth Euler accuracy and wave-speed ablation	Contours, error table, unit-speed ablation	proposed
Post-shock Burgers	Resolution after shock formation	Profile and reference metrics	proposed
Scaled Lax	Scale invariance, wave-speed behavior, and reference error	Normalized panels and profile	proposed
Blast wave	Unassisted failure and common-limiter robustness	Profile and robustness table	proposed
Shu–Osher	Shock–entropy resolution	Profile with matched zoom	proposed
2D Riemann	Resolution at equal mesh and equal DOFs	Separate contour and line-cut figures	proposed
KXRCF ablation	Threshold sensitivity and selective damping	Indicator/profile plot and activation table	proposed
Cadence ablation	Post-step versus stage-wise adapted OFDG	Error/runtime table	proposed
Strong scaling	Fixed-work scaling on one, two, and four ranks	Speedup and efficiency panels	proposed

The dry-run expansion additionally records the exact domain, initial and boundary data, final time, degrees, meshes, CFL rule, time integrator, methods, metrics, reference check, estimated cost, priority, and intended report artifact for every concrete row.

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Eidgenössische Technische Hochschule Zürich
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